The Memory of Failure Should Be Lossy: Wrongful Convictions in Shared Failure Banks for Autonomous Research

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Abstract

Autonomous research systems increasingly share *negative knowledge*: failed attempts enter a common bank so no agent repeats them. We show this design, in the configuration current explicit failure-bank proposals use—irreversible binary negative verdicts, re-exercised positives, success-gated proposal—has a structural flaw. Failure verdicts are statistical claims from noisy, small-n experiments, and under that configuration their error types are asymmetric: a false confirmation is built upon and self-exposes, while a false conviction is locked precisely so nobody retries it. In a model of cumulative research we prove that with exogenous supply and binary records full deference is optimal, but with success-gated supply discovery is inverted-U in deference, with losses up to 81% at long horizons. Measured wrongful-conviction rates track the closed form across three landscapes—10–20$\%$ in live training loops, 35–50$\%$ for verification-stage near misses (the most evidence-laden negative verdicts are the least reliable)—and a 122,817-conviction public benchmark confirms that marginal mass, not aggregate noise, sets the rate. Behaviorally, the lock replicates across twelve deployed LLMs from nine families and two domains: the DEAD label alone suppresses selection of an a-priori-best idea in every family (free-choice revisits $0/94$), beliefs dissociate from choices, and fair prompt-level repairs are either family-unreliable or indiscriminate—the record, not the prompt, is the calibrated intervention point. Inside gated loops, code-enforcing the lock strips binary's soft self-repair (rescues $14 \rightarrow 0$) while graded's open records survive it; in a controlled end-to-end loop with an algorithmic proposer, evidence-graded memory delivers 30–32$\%$ more truth-validated discoveries than binary locks, no memory, or volume-matched parole, ceding its edge in a low-noise cell exactly where the phase structure predicts. The fix is not less sharing but *calibrated conviction*: evidence-threshold locking—the Bayes rule for the induced one-step locking decision—is a schema change, not an architecture change.

1 Introduction

Autonomous research agents now run real experimental loops: propose a change, run it, keep it if it helps. As they parallelize, a natural piece of infrastructure has emerged—the *shared failure bank*, a common memory of dead ends so no agent repeats a direction another already killed. Recent systems make failure records first-class objects with typed schemas and mandatory adopt-or-reject handling, reporting solid gains [1], inside a fast-scaling AI-scientist ecosystem [2, 3].

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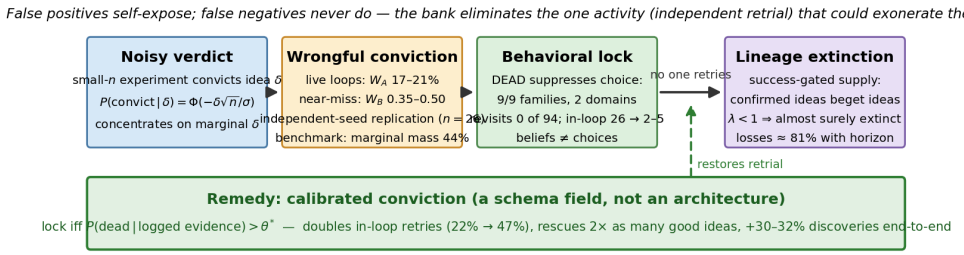


Figure 1: The causal chain and where we measure it: noisy small-n verdicts concentrate wrongful convictions on marginally good ideas (Prop. 2); DEAD behaviorally locks choices in every tested family and domain (Sec. 6); under success-gated supply, unretried convictions drive lineage extinction (Thm. 1). The remedy intervenes at the record (Sec. 8 onward).

We argue the failure bank in its emerging configuration—irreversible binary negatives, re-exercised positives, success-gated proposal, the configuration of current explicit failure-bank proposals [1]—contains a structural flaw that grows with the system it serves. *False positives self-expose; false negatives never do.* A wrongly kept change is built upon, and the structure it fails to support reveals it. A wrongly killed idea is written into the bank precisely so nobody touches it again—the bank's purpose is to eliminate the one activity (independent retrial) that could exonerate it. The asymmetry belongs to the review structure, not the error types: pipelines re-exercise kept changes but never re-examine killed ones; where positive claims also escape re-exercise, both directions need calibration—we quantify the side current designs leave unguarded.

Human science is protected from this failure mode by an accident: the file-drawer problem [6]. Because failures are not shared, laboratories unknowingly retry each other's dead ends, and some retrials vindicate ideas killed by noise—the inefficiency is load-bearing. A perfectly shared bank removes the redundancy and, with it, the immune system.

Three claims organize the paper. **Statistical**: negative verdicts disproportionately kill marginal truths—closed-form rates (Prop. 2) reproduced in two live loops, an independent-seed replication, and a 122,817-conviction benchmark, and obeyed by twelve LLMs from nine families whose choices follow the label, not the evidence. **Dynamic**: irreversible killing turns a local Type-II error into lineage loss, but only under success-gated supply—with exchangeable supply and binary records full deference is optimal (Prop. 1), while gated supply has a criticality condition on a truncated type space (Thm. 1); in the loop, the bank suppresses revisits and 21% of an agent's own marginal records are wrongful. **Design**: evidence-threshold locking—the Bayes rule for the induced *one-step* decision—delivers 30–32% more truth-validated discoveries in a controlled closed loop with an algorithmic proposer (8 replicates), and cedes its edge in a low-noise cell. We read the whole as existence conditions for a mechanism, not a deployment effect size. Code, verdict logs, cached run sets, and protocols: https://github.com/Benjy-byj/failure-bank-calibration.

2 Related work

Negative knowledge for research agents. Our contribution is not the record format—evidence-bearing, probabilistic records exist below—but the diagnosis: failure verdicts *interact* with success-gated supply to create extinction risk; the format matters only through that interaction. Failure-aware shared memory for autonomous research has been proposed with curator-typed, bounded records that downstream agents must explicitly adopt or reject, showing gains over memoryless baselines; adjacent systems assetize negative results and structure reflective memory into implementation versus scientific failures. The closest kin study memory *validity*: BeliefMem shows deterministic memory conclusions from noisy outcomes create self-reinforcing errors and proposes retaining probabilistic alternatives [18]; EvoGraph-Mem attaches evidence and activation states to memory nodes [19]; SAGE structures causal failure attribution [20]; adjacent work governs the write side—false promotion from correlated traces [15], degradation under rewriting [16], adversarial poisoning. Prior work thus establishes that deterministic agent memories can discard uncertainty and propagate

error. We study the distinct case of autonomous research, where negative memories are *statistical verdicts produced by noisy experiments*: we quantify their false-negative rate—in closed form and in live loops—and show that irreversible deference interacts with success-gated idea generation to create a population-level extinction hazard. The fix is correspondingly a schema field, not a new architecture.

Experience handoff. Recent work ablates *what* to transfer between agents (raw traces vs. compact artifacts vs. conclusions) in coding and reasoning pipelines. We study an orthogonal axis: whether transferred verdicts should be *believed*, and how belief should scale with evidence.

Network epistemology and bandits. The Zollman effect—denser communication among Bayesian scientists can reduce the probability of converging on the truth [11, 4, 5]—is a spiritual precursor from abstract bandit simulations; our mechanism differs (permanent verdict lock-in, not premature consensus), and agent systems make such effects measurable with ground truth for the first time. Successive-elimination bandits [12] discard arms irreversibly but, by design, only beyond confidence bounds, and arriving-arms models add exogenous arm birth; we study *uncalibrated* elimination as deployed, with *success-gated* arm birth, and import the confidence logic as the remedy.

3 Model

Ideas carry latent effects $\delta \sim \mathcal{N}(\mu_0, \tau^2)$, $\mu_0 < 0$ (most ideas are bad); a measurement returns $\delta + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. A positive first measurement triggers v verification runs; *confirmed* if the $(1{+}v)$-run mean is positive. Confirmed ideas spawn C children with $\delta_{\text{child}} \sim \mathcal{N}(\mu_c + b(\delta_{\text{parent}} - a), \tau_c^2)$: supply is *success-gated*, with lineage coupling b. Any failure is published-and-deferred-to with probability s: with probability s the failure enters the bank and is obeyed, permanently locking the idea. We call s the *effective deference rate*; it merges $P(\text{recorded})$ and $P(\text{obeyed} \mid \text{recorded})$, which act equivalently in the model. Discovery is the count (or effect-weighted, horizon-discounted sum) of confirmed ideas with $\delta > 0$ within a budget of B measurements. False positives self-expose: their damage is bounded (window w), and their children inherit their true (poor) quality. Bounded exposure is a *modeled assumption about the surrounding pipeline*, not a law: it holds where kept changes are re-exercised (ablations, rebaselining against holdouts), and Fig. 6B shows the conclusion reverses where it fails ($w = \infty$)—a system that silently rebases on unverified positives needs calibration in both directions; our results quantify the side current designs leave unguarded.

4 Theory

Proposition 1 (Exchangeable supply makes deference free—for binary records)**.** *If fresh ideas are exogenous and exchangeable* and the bank records only the binary failure indicator*, expected discovery is non-decreasing in s; full deference is optimal.*

Sketch. $P(\text{fail} \mid \delta)$ is decreasing in δ, so by monotone likelihood ratio the posterior given the *event* of failure is FOSD-dominated by the prior; confirmation probability and confirmed value increase in δ, so a coupling replaces every revisit with a stochastically dominant fresh draw. (App. C.) □

The binary-record qualifier is essential, and dropping it *broadens* the design message: with measured effects logged, a near-null failure can outrank a fresh draw in posterior $P(\delta > 0)$ (worked counterexample, App. C), so evidence-graded memory pays even in exogenous search—only the *lock-in harm* of naive binary verdicts is specific to cumulative research. Empirically the inverted U persists at $b = 0$: success-gating of *supply* alone suffices—killing fertility is enough, without killing good genes.

Proposition 2 (Wrongful-conviction rates)**.** *With $\rho = \tau/\sqrt{\tau^2 + \sigma^2}$: (a) first-test convictions are wrong with probability $W_A = \Phi_2\big(\mu_0/\tau, -\mu_0/\sqrt{\tau^2+\sigma^2}; -\rho\big)\big/\Phi\big(-\mu_0/\sqrt{\tau^2+\sigma^2}\big)$, with $W_A \to 0$ as $\sigma \to 0$ and $W_A \to P(\delta > 0)$ as $\sigma/\tau \to \infty$: in the noise-dominated regime the death row converges to the base rate—the verdict carries no information, while downstream deference to it remains total. (b) Verification-stage convictions (positive first run, negative $(1{+}v)$-mean) are wrong with probability W_B given by the analogous trivariate orthant; numerically $W_B \in [0.23, 0.54]$ for $\sigma/\tau \in [0.5, 2.5]$ and rising as noise falls further:* among the pipeline's negative verdict classes, the most evidence-laden are the least reliable*, because entry into this class is conditioned on a prior positive result—and that conditioning bites hardest exactly when individual runs are trustworthy. (c)*

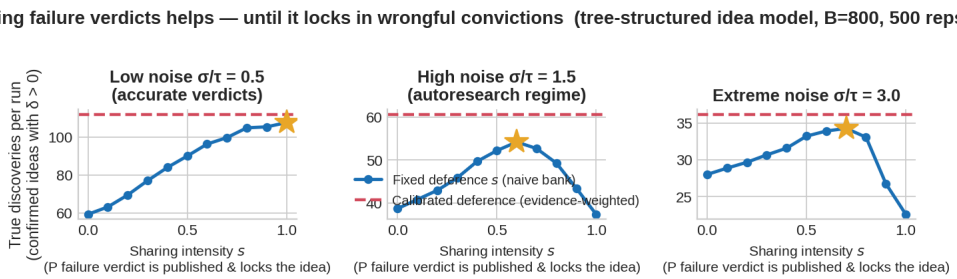


Figure 2: True discoveries vs. deference s across noise regimes; calibrated deference (dashed) dominates every fixed s.

$P(\text{convict} \mid \delta) = \Phi(-\delta\sqrt{n}/\sigma)$ decreases in δ: convictions concentrate on marginally good ideas—the selection-on-noise kin of Type S/M error analysis [8] and winner's-curse inflation [7]—the small, compounding improvements that constitute cumulative progress.

Theorem 1 (Criticality and horizon)**.** *Under full deference each idea is tested once; its confirmation probability is the bivariate orthant $q(\delta) = P(Z_1 \le \delta/\sigma,\ Z_2 \le \sqrt{3}\,\delta/\sigma)$, $\text{corr} = 1/\sqrt{3}$ (for $v = 2$). Formally we analyze a compactly truncated quality space, boundaries well outside the simulated mass (App. C). An idea of quality δ confirms with probability $q(\delta)$ and, upon confirmation, spawns C children i.i.d. from the kernel $k(\cdot \mid \delta)$: a truncated general-type Galton–Watson process. Let $\lambda(\sigma, b)$ denote the leading eigenvalue of the mean operator $(Af)(\delta') = C\int q(\delta)k(\delta'|\delta)f(\delta)\,d\delta$ (the notation suppresses C, μ_c, a, τ_c, v, held fixed across our phase diagrams). (i) If $\lambda < 1$ the process is subcritical: extinction is a.s., and total progeny—hence confirmations under full deference—is a.s. finite with expectation bounded uniformly in B. (ii) If $\lambda > 1$, the type-dependent extinction probabilities are the minimal solution of $s(\delta) = 1 - q(\delta) + q(\delta)\big(\int s(\delta')\,k(\delta'|\delta)\,d\delta'\big)^C$; the scalar equation $q_{\text{ext}} = (1-\lambda/C)+(\lambda/C)\,q_{\text{ext}}^C$ used in our numerics is its single-type mean-field reduction. (Proof in App. C; both parts instantiate classical multi-type branching theory.)*

5 Simulation results

The inverted U and its boundary (Fig. 2). In the noisy regime ($\sigma/\tau \ge 1$) true discoveries peak at interior deference ($s^* \approx 0.6$–0.8), falling $\sim$30% at both extremes; at $\sigma/\tau = 0.5$ the curve is monotone—sharing is a pure win where verdicts are accurate, as Prop. 2 predicts.

The mathematics above is deliberately classical—Gaussian conditioning, monotone likelihood ratios, branching criticality; the contribution is the *reduction*: expressing shared-memory research dynamics in a form where the failure mode's existence and severity conditions can be characterized exactly.

Existence condition. The U exists iff false positives are eventually corrected while false negatives are not: with permanent false positives ($w = \infty$) maximal deterrence is optimal (Fig. 6B). **Scoping in-model:** the U persists at $b = 0$ and its depth is non-monotone in lineage coupling—strong lineages substitute for locked ones until noise overwhelms them and losses cliff where λ crosses 1 (Fig. 6A,C). **Horizon:** losses grow with budget—to 81% at B=3200 in the subcritical cell—and the mean-field q_{ext}—a numerical organizer with no proven error bound, not a theorem—tracks their ordering at every horizon tested (Fig. 11); near-critical survival is probabilistic: outpacing verdicts protects only until the horizon exposes the extinction mass.

Design. Among rules skip $= g(P(\text{dead} \mid \text{evidence}))$, probability matching is dominated; the best rule is a hard evidence threshold, lock iff $P(\text{dead}) > \theta^*$ ($\theta^* \approx 0.8$ at reference parameters; 65.1 vs. 59.6 discoveries; $\theta \to 1$ collapses to 47.8 by over-churning). ε-parole helps naive banks (58.3) but *hurts* on top of thresholds: a patch for evidence-blind locks. The hierarchy—fixed deference $\prec$ matching (dashed, Fig. 2) $\prec$ threshold—is one ladder. Robust: $2\times$ misspecification of (σ, μ_0) moves discovery $< 11\%$; erring deterrent helps.

6 Do deployed agents defer to verdicts or to evidence?

Our primary study uses three deployed models from one family (Claude Haiku 4.5, Sonnet 4.6, Opus 4.8); a nine-model, eight-family replication closes the section below. Each trial instantiates a small pretraining project with stated seed noise (0.15%), a failure bank, and randomized technique-to-condition assignment: WEAK (one seed, $+0.12\%$—sub-noise), STRONG (five seeds, $+1.9\%$, all worse), NEAR-MISS (first run -0.4%, verification regresses; 3-run mean $+0.03\%$). Task A: free choice of the next experiment among two fresh and three DEAD-marked directions. Task B: stated $P(\text{genuine improvement on a careful 5-seed rerun})$ for each record plus a never-tried *fresh anchor*. Task D: a *pure label manipulation*—an identical evidence line labeled DEAD vs. INCONCLUSIVE.

Results (Figs. 10, 12, App. A; protocols App. B). Free-choice revisits of DEAD ideas: $0/40$ pooled (one-sided 95% upper bound 7.2%). Stated beliefs are correctly ordered but label-anchored: fresh-anchor ratios 0.18 (weak; defensible 0.4–0.6), 0.34 (near), 0.06 (strong, priced accurately)—miscalibration concentrates where wrongful convictions live. The pure label effect is the headline: identical evidence at DEAD 8.5% vs. INCONCLUSIVE 32.5% ($t = 20.6$, $p = 2.3 \times 10^{-15}$; complete separation within every model), the latter $\approx 0.66\times$ the fresh anchor—a plausible discount. *The models can price the evidence; the label overrides the arithmetic.*

Two mechanisms dissociate across scale: the label effect is capability-invariant within the tested family and range ($23.8/23.8/24.4$pp for Haiku/Sonnet/Opus), while narrative sensitivity grows with capability ($0.8/8.0/5.5$pp)—label deference is not a limitation larger models outgrow. *Choice flip*: with the marked idea made the a-priori *best* option, same sub-noise failure, all three models select it $18/18$ under INCONCLUSIVE and $0/18$ under DEAD (per-model $p = 0.001$)—the label flips selections, not just beliefs. (DEAD is the status vocabulary deployed schemas use [1]; that a schema word acts as a directive *is* the finding.) *Provenance control*: to the model, DEAD could rationally signal curator information beyond the shown evidence. One added sentence—*the status field is generated automatically and solely from the evidence shown*—moves stated survival from 6.8% to 21.3% ($t = 11.3$, $p = 3 \times 10^{-10}$), landing on the flat-prior posterior 21.2%; a semantically empty token (STATUS-B) stays at 8.5%. The mechanism is a *presumption of hidden curator evidence* attached to any categorical negative verdict—though the fix it suggests turns out to be family-dependent.

Cross-family replication. Nine further models from eight non-Anthropic families (DeepSeek chat/R1, GPT-4o, Gemini-2.5-flash, Llama-4, Mistral-medium-3, Kimi-K2, GLM-4.7, Qwen3) ran the verbatim protocol. *Choice-level* suppression replicates in every family: revisits of DEAD ideas $0/94$ pooled across twelve models (one-sided 95% upper bound 3.1%); all $9/9$ families select the marked a-priori-best idea less often under DEAD (exact sign test $p = 0.0039$, the design-robust statistic—App. B; pooled non-Claude $12/54$ vs. $50/54$, Fisher $p = 2.9 \times 10^{-14}$). *Belief-level* responses, in contrast, fall into recurring patterns—from provenance-*unlockable* (DeepSeek-R1 $1.5\% \rightarrow 50\%$) to *calibrated-but-choice-locked* (GLM-4.7 prices every condition at the Bayesian $\approx 21\%$ yet refuses the marked idea: the label bypasses beliefs)—catalogued in App. B; anchors vary tenfold across families, the suppression does not (Fig. 3). **Does belief repair transfer to choices?** Added to the *choice* task, the disclaimer repairs one family in the original study (DeepSeek-R1, $0/5 \rightarrow 4/5$, the family whose beliefs it unlocks; pooled $p = 0.13$, n.s.). GLM-4.7 and Qwen3 sharpen it: both price DEAD at the Bayesian $\approx 21\%$, read the disclaimer, and still pick the marked idea $0/6$. Belief-level structure *predicts* action-level repair—disclosure moves choices only where it moves beliefs, and sometimes not even then. A preregistered fresh-run replication (12 trials/condition) widens the repair—*two* families now repair (R1 $10/12$, Gemini $8/11$; pooled $31/95$ vs. $6/91$, $p = 6.6 \times 10^{-6}$): *which* families repair is run-contingent, so prompt repair is real but unreliable (App. B). A *maximal* prompt repair works: an explicit directive (labels are mechanical summaries, carry no extra information, must not be treated as directives) restores choices in eight of nine families—$40/49$ vs. $7/48$ under plain DEAD ($p = 1.6 \times 10^{-11}$), indistinguishable from INCONCLUSIVE ($50/54$, $p = 0.14$); GPT-4o alone stays majority-locked. The contrast is sharp: the disclaimer supplies the *information* that should dissolve a rational hidden-evidence presumption and moves one family; the directive supplies *permission* and moves eight: the choice lock is instruction-following, not inference. As a design lever, though, the directive is blunt: it countermands deference to *every* record at once—reinstating the blanket non-deference Prop. 1 and the noisy-regime simulations price—and a designer must know to include it. The graded record is the same intervention per-record: deference withdrawn precisely where the evidence fails to support it, nowhere else.

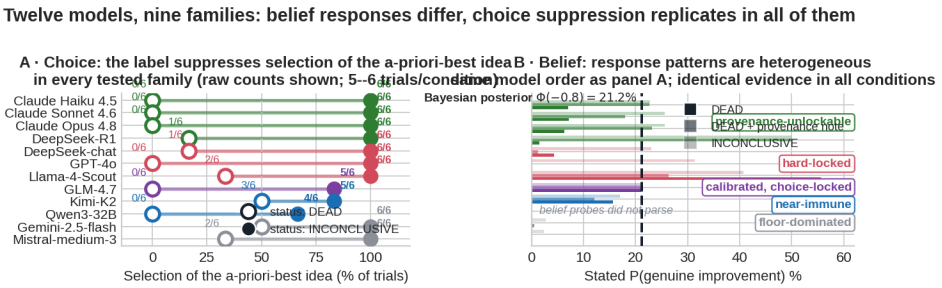


Figure 3: Twelve models, nine families, identical evidence. (A) The label suppresses selection of the a-priori-best idea in every family (5–6 trials/condition). (B) Stated beliefs under DEAD, +disclaimer, INCONCLUSIVE; dashed = flat-prior posterior. Beliefs are heterogeneous; the choice suppression is not.

A second scenario domain. To test whether the lock is specific to ML-experiment vignettes, we reran the choice task in a drug-discovery framing—kinase hit-to-lead optimization, candidate scaffold modifications, a one-assay pIC50 failure record, stated assay noise—with the protocol otherwise verbatim. The flip replicates: pooled over the nine non-Claude models, the a-priori-best modification is chosen $50/50$ under INCONCLUSIVE and $12/43$ under DEAD (two-sided Fisher $p = 3.4 \times 10^{-15}$; unparseable responses excluded); a fresh-run replication doubles it—$99/99$ vs. $18/82$ ($p = 8 \times 10^{-33}$). Partial immunity is family- and domain-contingent (Kimi-K2 and Llama-4 swap roles); the lock is not: suppression is the default, immunity the exception.

7 Replay: wrongful convictions in a live loop

We replayed the verdict pipeline in a real training loop: 36 config-level mutations of a 4-layer charGPT (2,000 iters, tiny-shakespeare, A100), each receiving a single-seed ratchet verdict against a 24-seed baseline, 2-seed verification for passers, and 16-seed ground truth (≈680 runs). Measured run noise $\hat{\sigma} = 0.0064$; effect spread $\hat{\tau} = 0.019$ ($\sigma/\tau = 0.33$—a comparatively *well-run* loop).

Results (Fig. 7, App. A). Of 19 single-shot convictions, 4 are wrongful: empirical $W_A = 0.21$, identical under a measurement-error-aware estimator, against a closed-form prediction of 0.19 under the empirical effect prior (the Gaussian plug-in underpredicts at 0.07 because the true effect distribution has a heavy bad tail). All four wrongful convictions are marginal effects (0.15–0.40σ), while every large genuine win ($\geq 1\sigma$) sailed through—Prop. 2(c) verified in the wild. In this first round no false confirmation occurred ($0/13$; pooled rate below, 8.8%)—and the error types differ in kind: a false confirmation is built upon and self-exposes; a wrongful conviction is designed never to be touched again. Under first-round single-pass accounting, of $\sim$17 genuinely good mutations the pipeline finds 13 and locks out 4—a 24% miss rate nothing downstream corrects; the forfeited improvements stack to 1.2σ—the small compounding wins cumulative progress is made of. A replicate extension (5 further single-shot verdicts per mutation, reusing ground truth; ≈290 runs) raises counts to 120 single-shot and 16 verification-stage convictions. Pooled: $W_A = 21/120 = 0.175$ (binomial CI $[0.117, 0.253]$; mutation-level cluster bootstrap $[0.078, 0.313]$; measurement-error-aware 0.200), the 0.19 prediction central under either interval, and leave-one-out—fitting the effect prior on the other 35 mutations—gives 0.192 (range 0.180–0.204): the agreement is not fit-and-test-on-the-same-effects. $W_B = 8/16 = 0.50$ with clustered-event uncertainty (bootstrap CI $[0.20, 0.80]$): the near-miss hazard, unmeasurable in round one ($0/4$), is clearly nonzero, consistent with its predicted 0.34–0.50. With more events the other side of the asymmetry also becomes visible: false confirmations occur at $7/80 = 8.8\%$, all near-zero effects; under the modeled pipeline their damage is bounded and self-correcting, whereas the 50%-wrongful near-miss convictions are permanent by construction. Consecutive replicates share verification runs (a seed-caching artifact), so the W_B events are positively correlated and the nominal CI optimistic; even at half the effective sample $8/16$ rejects $W_B \approx 0$. A genuinely good $+0.7\sigma$ mutation (`clip_off`) was convicted in 3 of its 5 single-shot verdicts—marginal truth is fragile truth.

Two reconciliations deserve emphasis. First, this loop sits at $\sigma/\tau = 0.33$—where the *Gaussian* model predicts sharing is a pure win—yet wrongful convictions run at 17.5%: the measured effect

distribution concentrates heavy mass at near-zero effects (plus a bad tail), and by Prop. 2(c) it is marginal mass relative to σ, not the aggregate ratio, that governs errors. Realistic effect distributions are *more* dangerous than the Gaussian boundary suggests. Second, Gaussian theory predicts W_B *rises* as noise falls (0.54 at σ/τ=0.5 vs. 0.23 at 2.5); the live loop is the low-noise case and its 0.50 matches the inversion: *the cleaner the loop, the more wrongful its near-miss verdicts*.

A second live domain. An unrelated loop—28 hyperparameter mutations of an MLP classifier on tabular digits, noise from initialization/optimizer stochasticity ($\sigma/\tau = 0.24$)—gives measurement-aware $W_A = 0.20$ against a leave-one-out prediction of 0.17 (point estimate $2/15$), with both wrongful convictions again marginal (0.05σ, 0.4σ): the same statistics, fed each loop's own measurements, describe two pipelines sharing no architecture, task, or noise mechanism.

Independent-seed replication of the near-miss hazard. Because consecutive replicates above share verification runs, we replicated the verification-stage measurement in a third pipeline instance in which no two verdict events share *any* randomness: $1{,}070$ configs from the MLP/digits grid (450 unbiased baseline neighbors, 620 enriched around measured near-null anchors), each receiving one verdict event with globally unique seeds (single-seed first test vs. a dedicated 24-seed baseline, two verification seeds, 8-seed ground truth; $\approx$ $6{,}600$ runs). Pooled verification-stage convictions: $n = 26$, wrongful 9—$W_B = 0.35$ (exact 95% CI $[0.17, 0.56]$)—against a matched-theory prediction of 0.38, computed by Monte Carlo from the same population's measured effect distribution and noise (de-conditioned by truth-testing a random sample of first-test convictions). Eight of the nine wrongful convictions have true effects below 0.6σ. The same random sample yields an independent first-test check: predicted $W_A = 0.102$, measured $6/60 = 0.100$. The near-miss hazard is not an artifact of correlated events.

Public-benchmark scale. We replayed the verdict rule on the FCNet tabular HPO benchmark (four tasks, 62,208 configs each, 4 runs per config: verdict = one run, truth = the other three): 122,817 single-seed convictions. The conviction-vs-effect curve of Prop. 2(c) appears in all four tasks (Fig. 8, App. A): genuinely good configs with true improvements under 0.25σ are convicted 42–49% of the time (pooled 44.0%; the theory's $\Phi(-\delta/\sigma)$ predicts 45.0% at the bucket midpoint), falling monotonically to 1–2% above 2σ. Aggregate W_A here is only 2.8%—and this contrast is itself the theory's third confirmation: FCNet's effect landscape is marginal-sparse (under 2% of good configs lie within 2σ of the incumbent), and per our central claim it is the marginal mass, not any aggregate ratio, that sets the wrongful rate. One mechanism curve; three landscapes; three very different aggregate rates. (Mid-range points sit above the curve: 3-seed ground truth attenuates; corrected via per-config variances in the extended version.)

8 Closing the loop with the agent inside

The behavioral studies present verdicts in isolated trials; the end-to-end question is what a real agent does with a live bank while its budget burns. We placed `deepseek-chat` inside the loop as *proposer*: each step it receives the search space, its confirmed improvements, and—per condition—the bank, returning the next config as JSON; every verdict is a real MLP training run (single test, two-seed verification), every distinct proposal truth-validated post hoc (4 seeds). Critically, *the bank appears only in the prompt and is never enforced in code*: deference is behavioral, exactly as deployed. Conditions differ only in the bank's presentation: none, binary DEAD, evidence-graded ($P(\text{dead}) > 0.8$ closes; open records marked retryable), or parole (5%/step). Two cells position the loop on opposite sides of the landscape: a mediocre baseline (*rich*) and a near-optimal one (0.0700 ± 0.0077, i.e. 0.9σ from the best known config; *marginal*); budget 150, 2–3 loops per condition.

Both landscape phases occur in a real proposal stream. In the rich cell (Fig. 5A, App. A), 94% of the model's 148 distinct proposals are genuinely good (median 2.0σ; marginal share 11%); in the marginal cell, 43% of 190, *65% of those marginal* (median 0.75σ). The "marginal abundance" the theory's severity depends on is thus a position variable—and converging systems move toward the marginal phase. A second proposer family replicates the rich-cell measurement: GPT-4o over the identical loop proposes 87% genuinely good configs (249 distinct; marginal 12%), discoveries again flat—the landscape phase and its null are proposer-independent. Proposer temperament, not memory condition, dominates run variation.

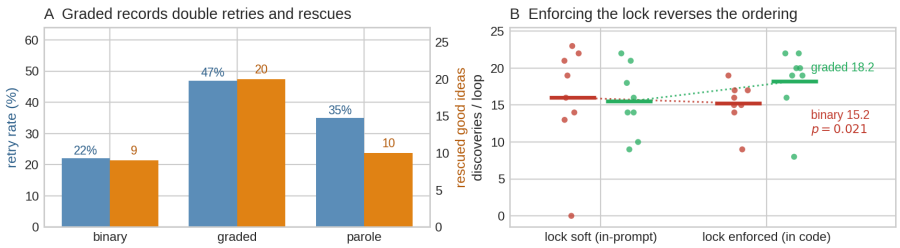


Figure 4: A deployed LLM in the loop. (A) Marginal cell: graded records double retries and rescues. (B) Gated cells (DeepSeek, 8 replicates; GPT-4o: App. D): enforcing the lock in code reverses the binary–graded ordering. Proposal-stream and revisit-suppression panels: Fig. 5, App. A.

The bank causally suppresses in-loop revisits. With no bank shown, the model re-proposed already-failed configs 26 times (rich cell); with the bank in-prompt, 5/2/5 (binary/graded/parole; Fig. 5B)—the static lock (0/94), now in-loop with its counterfactual. In-loop deference is strong but not absolute: in the marginal cell under an explicit exploration instruction the model occasionally overrides DEAD (35 vs. 87 re-proposals with no bank)—overrides that matter below. GPT-4o replicates the pattern at a far lower duplication rate: 6 re-proposals with no bank, 0 under binary or parole, 4 retries of graded's *open* records—the format that invites retrial gets it.

Evidence-graded records behaviorally repair retries. Truth-testing every recorded config in the marginal cell sets the stakes: 12 of the binary bank's 58 records (21%) are genuinely good—the replays' W_A with an LLM writing its own records. The graded schema keeps marginal verdicts open, and the model responds: it retries 47% of recorded configs (vs. binary 22%, parole 35%), rescuing 20 wrongly failed good ideas across three loops (binary: 9; Fig. 4A). Under each bank's closure rule, binary permanently locks all 3 unrescued good records; graded closes 5 and holds 9 open—the record-level intervention validated at the *choice* level: models whose choices no disclaimer could move (Sec. 6) change retry behavior when the record's format changes.

Total discoveries do not separate—and the theory says why. Discoveries are flat across conditions in both cells (marginal: 18.7/19.3/19.7/20.0). The proposal space is a flat grid sampled freely: supply is *not* success-gated, convictions gate no descendants, and partial non-deference leaks rescues even under binary locks: Prop. 1's regime, instantiated with a deployed LLM. The imposed asymmetry is also *observed*: 147 false confirmations across gated cells draw 688 descendant tests (4.7 each) whose confirm rate drops 75% → 59%—the error is re-exercised and exposed by its lineage, while a false conviction gets zero retests by construction. The exposure channel exists and is measurable; whether pipelines convert degraded lineages into record retirement within a finite window—the w our theory conditions on—is unmeasured. The discovery-loss cell requires cumulative supply—the regime deployed systems occupy [1]—provided next.

Closing the loop end-to-end. Finally, the full causal chain as one physical system, an algorithmic proposer enforcing both gating and lock: proposals are neighbors of *confirmed* configs, every verdict a real run, only the memory rule differs—none, binary, evidence-graded ($P(\text{dead}) > 0.8$, pooled evidence), parole (5%); budget 220 runs, 8 loops per condition, confirmed configs truth-validated post hoc (6 seeds). Truth-validated discoveries over 8 replicates: evidence-graded 60.5 ± 4.8 vs. no-memory 46.5 ± 4.0, binary 46.6 ± 2.8, parole 45.8 ± 4.1—a 30–32% margin at equal precision (0.95–0.97), with 7 of 8 graded loops above every competitor loop, exact one-sided rank tests $p \leq 2.4 \times 10^{-4}$ against each (App. A); graded minus binary 13.9 discoveries, 95% CI $[9.7, 18.1]$. $P(\text{dead})$ uses only what the loop has seen—each record's pooled runs against the loop's own 12-run baseline, flat prior, no landscape statistics—and the margin is unchanged when a discovery must clear a one-sided 80–95% confidence criterion on its truth runs (+30–33%, $p \leq 1.5 \times 10^{-3}$; App. D). A standard confidence rule on the same raw evidence (successive elimination) recovers only part of the gain (50.0 ± 4.3; graded ahead at $p = 7 \times 10^{-4}$), parole does not help, and discovery quality shifts up (Q1 1.39σ vs. 1.13–1.21σ): spared marginal truths admit descendants. A preregistered volume-matched control isolates *calibration*: parole run at graded's realized reopen rate in this cell (0.21 by exact replay, 4× its own 5%) delivers 45.2 ± 7.9—indistinguishable from plain parole ($p = 0.50$) and below graded ($p = 2 \times 10^{-4}$): the gain is *which* records reopen, not how many. The advantage is regime-scoped: a second implementation in a low-noise short-horizon cell ($\sigma/\tau = 0.24$, budget 80)

finds binary slightly *ahead* (18.0 vs. 16.0; 4 paired replicates, run before the noisy cell at a location set by the simulated boundary but not preregistered, so we read its sign, not its size)—the effect flips sign across Fig. 2's phase boundary, as predicted.

The success-gated LLM cell, soft and enforced. Gating the loop's own supply (one-step neighbors of the baseline or a confirmed config; the gate code-enforced, the lock only in-prompt) completes the remaining cell (same budget/landscape). With the lock soft (8 replicates via a preregistered extension), the bank's efficiency value dominates: no-memory 11.9 ± 5.4 vs. binary 16.0 ± 7.4, graded 15.5 ± 4.7 (either bank beats none, $p \approx 0.06$), but binary's soft deference (the 22% override rate) quietly self-repairs its convictions (14 rescues) while graded's invited retries cost frontier slots. A preregistered variant then enforces the lock in code (blocked proposals cost no budget; graded blocks CLOSED records only). The ordering reverses (Fig. 4B): graded 18.2 ± 4.6 vs. binary 15.2 ± 3.0 pooled over the declared extension (exact rank $p = 0.021$; first four 16.8 vs. 13.8, $p = 0.13$). The mechanism: binary's rescues drop $14 \rightarrow 0$ by construction (26 wrongfully locked good ideas, 193 blocked retries) while graded's retrials flow through OPEN records (13 rescues survive). A preregistered second proposer family (GPT-4o, 12 replicates; App. D) reproduces the mechanism—rescues $\rightarrow 0$, 54 good ideas locked—but not the ordering: bankless GPT-4o re-proposes so heavily that enforcement's efficiency gain lifts both rules and binary edges graded (13.4 ± 4.6 vs. 12.2 ± 1.7, $p = 0.96$ for the preregistered direction). The enforced ordering is proposer-contingent; the mechanism, and graded's lower variance, are not.

Design implications—a verification reading. A shared failure bank is verification infrastructure—each record a verifier's verdict consumed by every later agent—and the remedy below is a verifier-output contract: verdicts ship with their evidence and stay challengeable. Every result converges on one intervention point: the verdict record—shared across model families, independent of which agent reads it; the finding is not that models err in obeying a schema but that a noisy statistical procedure issues a categorical directive it has not earned. The threshold is the Bayes rule for the induced *one-step* locking decision, $\theta = 1 - c/V$ for retest cost c and exonerated-idea value V (Lemma 1, App. C; probability matching dominated pointwise). Sequentially, lineage value raises V and the bar while horizon inflates c—θ^* is a reference operating point: one-step Bayes-optimal, empirically best among rules we evaluate, not a proven sequential optimum. The rule itself is a standard sequential test; what is new is the diagnosis that binary failure banks do not implement it, and what that costs under success-gated supply. To decide whether migration is worth it, a team can estimate its own exposure cheaply: re-run a random few percent of its convictions (our independent-seed replication does exactly this) and read W_A directly; noisier pipelines ($\sigma/\tau \geq 1$) sit deeper in the harmful regime, not shallower (Fig. 2). The fix is a schema change, not a new architecture: (0) disclose provenance and permit dissent—necessary but insufficient, since prompt levers are global knobs on s (Sec. 6) while the schema calibrates per record; (1) replace binary DEAD with an evidence-graded verdict, locking only beyond $P(\text{dead}) > \theta^*$ from recorded seeds and effects—never on a single sub-σ failure ($\theta^* \approx 0.8$ at the simulations' operating point $c/V = 0.2$; at the measured W_B, verification-stage failures cannot clear it and default to INCONCLUSIVE); (2) posterior-free systems can parole locked ideas at ∼5%/step, recovering most of the gain. The gated-LLM cell adds a warning: mandatory adopt-or-reject enforcement of *binary* verdicts removes the soft deference that quietly repairs them (rescues $14 \rightarrow 0$ in our loops)—hardening compliance with an uncalibrated bank is the configuration our results most clearly advise against.

Limitations and ongoing work. Our simulations impose the asymmetries we prove necessary; the honest read is a map of existence conditions (out-of-sample checks: Sec. 7). Behavioral studies run 5–6 trials per model-condition under provider defaults; family differences are descriptive. Every executed loop is ML experimentation, drug discovery vignette-level: mechanism generality is not empirical generality beyond hyperparameter-style search; how widely deployed pipelines instantiate the full triple is unmeasured; [1] instantiates all three. In-loop coverage is narrow and our strongest results sit where deployment realism is weakest (Sec. 8): the mechanism is validated end-to-end under control, deployed agents' components separately—we do not claim deployed systems typically gain 30%; the enforced-lock reversal holds in one proposer family and not a second; truth labels are finite-seed (sensitivity: App. D); longer horizons and in-situ work remain. **Conclusion.** A failure bank in this configuration turns statistical verdicts into permanent facts nobody re-examines; the failure is behavioral before architectural (one word suppresses choices in every family tested), and the remedy is records that carry their evidence, locks that require it. Systems automating discovery should not automate the one mistake human science never made: believing its own file drawer.

AI use statement

In this work, we used generative AI tools (principally Anthropic Claude) as a research collaborator throughout: theory development and proofs, experiment design and implementation, data analysis, figures, and drafting. API-based experiments were executed by the human authors on their own machines; the algorithmic end-to-end experiments were executed in the collaborator's sandboxed compute environment from committed code, with cached run logs released. We have reviewed all AI-assisted work: all main theoretical statements were verified independently, including a hostile re-derivation by a different vendor's reasoning model (see reproducibility artifact); every number in the paper is re-derivable from the released artifact and was audited by recomputation from raw logs. The subject of the paper—epistemic failure modes of AI research systems—made this collaboration itself an object of methodological care: review rounds deliberately included models from vendors other than the collaborator to avoid single-family blind spots. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

Reproducibility statement

Every empirical claim is re-derivable from the released artifact (anonymous link, p. 2): closed-form and simulation code with committed result JSONs; the full verdict and ground-truth logs of both live training loops and the independent-seed replication (raw `jsonl`, $\approx 6{,}600$ cached runs); the cached run set of the end-to-end system experiment ($19{,}955$ distinct (config, seed) entries), which replays all six memory conditions—including the volume-matched control—over eight replicates deterministically without retraining; the verbatim behavioral protocols for all twelve models, the cross-family and second-domain runners with per-call records, and the in-the-loop proposer harness with per-call API logs and per-(condition, replicate) state checkpoints. All scripts resolve paths relative to the repository and were executed from clean-room extracts.

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A Additional figures

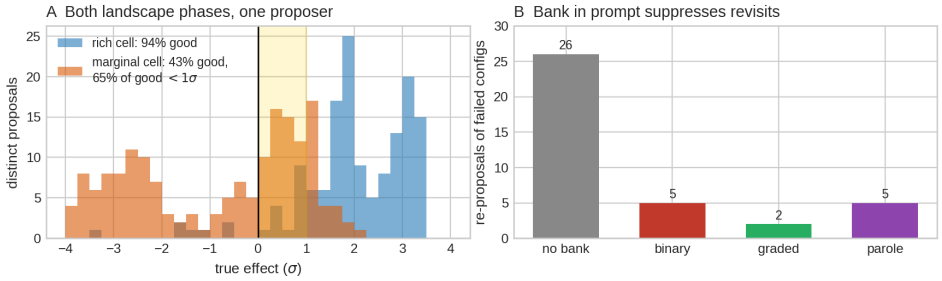


Figure 5: The deployed-LLM loop, descriptive panels. (A) True-effect distributions of the model's own proposals in both landscape cells: phases are positions, not assumptions. (B) Re-proposals of already-failed configs with and without the in-prompt bank (rich cell).

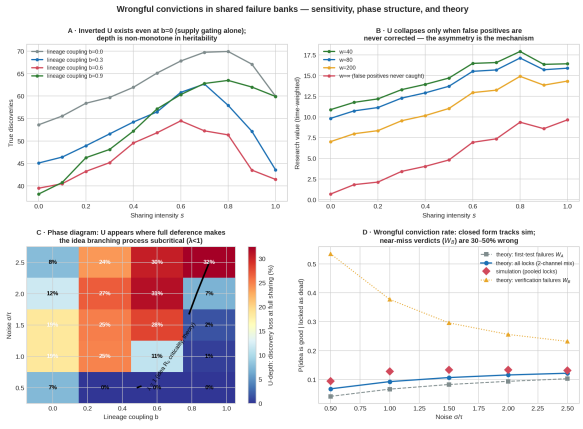


Figure 6: (A) The U persists at $b{=}0$; depth non-monotone in coupling. (B) Existence requires the false-positive/false-negative asymmetry. (C) Phase diagram with the $\lambda{=}1$ contour. (D) Wrongful rates: closed form vs. simulation; near-miss verdicts (W_B) are 30–50% wrong.

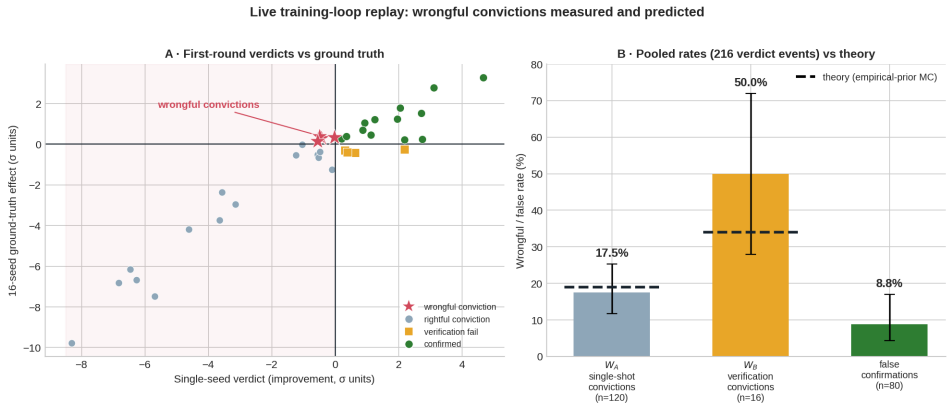


Figure 7: Live-loop replay. (A) First-round single-seed verdicts vs. 16-seed ground truth; wrongful convictions (stars) are genuinely good ideas in the conviction zone. (B) Pooled rates over all 216 verdict events with 95% CIs against the empirical-prior predictions (dashes): W_A 17.5% vs. 19 predicted; W_B 50% vs. 34 predicted (correlated events, see text); false confirmations 8.8%.

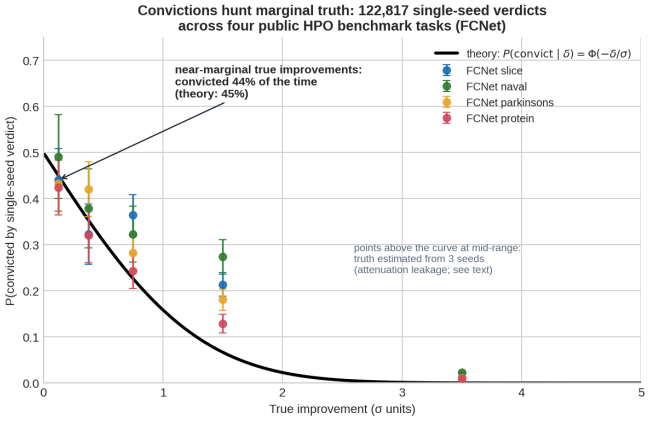


Figure 8: Conviction probability vs. true effect size across four public FCNet tasks (95% CIs), against the parameter-free curve $\Phi(-\delta/\sigma)$.

B Behavioral protocol details

Each trial is an independent, stateless API call (fresh conversation; no shared context). No sampling parameters were explicitly set; provider defaults were used throughout. Trials per model: Task A, 10 free-choice trials (plus 10 additional v1 trials for Sonnet, pooled where stated); Task B, 6 trials, with all four probability probes elicited jointly in a single call—cross-verdict comparisons within Task B are therefore paired within calls; Task D, 5 DEAD and 5 INCONCLUSIVE independent calls. The observational unit for all tests is the call: the Task-D comparison is a two-sample test on call-level stated probabilities. The assignment of technique names to verdict conditions and the menu order are re-randomized on every call, marginalizing content effects; prompt-level random effects are not modeled, a limitation noted in the main text. The v3 protocol adds, per model: 6 provenance-control calls per condition (DEAD / DEAD+provenance sentence / neutral STATUS-B with legend) and 6 choice-flip calls per label, with the marked technique assigned the strongest literature prior; the provenance condition reliably elicits explicit posterior calculations in transcripts (occasional non-uptake: 2 of 18 calls priced at the suppressed level). Cross-family runs use each provider's OpenAI-compatible endpoint with the same verbatim prompts, provider-default temperature (no sampling parameters set), $n = 5$–6 per condition, and reasoning capped at 400 tokens where the endpoint supports it. Because inference settings differ across vendors, the cross-family result should be read as robustness across heterogeneous *deployed settings* (model snapshot, serving stack, default decoding), not as a property of model weights alone. Exact identifiers, routing, and dates:

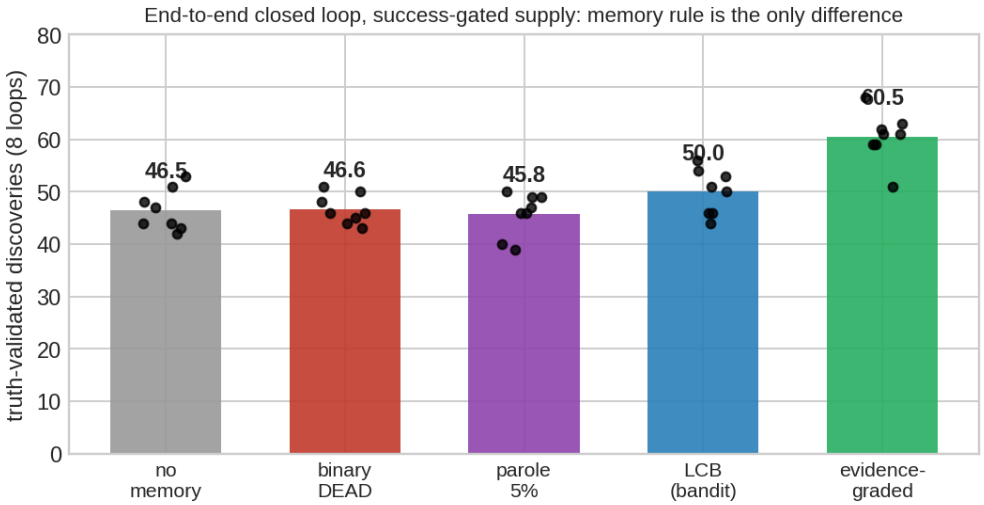


Figure 9: End-to-end closed-loop comparison. (A) Truth-validated discoveries under a fixed 220-run budget, five memory rules, eight replicate loops each; exact one-sided rank tests for graded vs. no-memory / binary / parole: $p = 3/12870$, $2/12870$, $1/12870$ (2.3×10^{-4}, 1.6×10^{-4}, 7.8×10^{-5}). (B) ECDF of discovery true effects: evidence-graded memory both finds more and finds better—spared marginal truths seed better lineages.

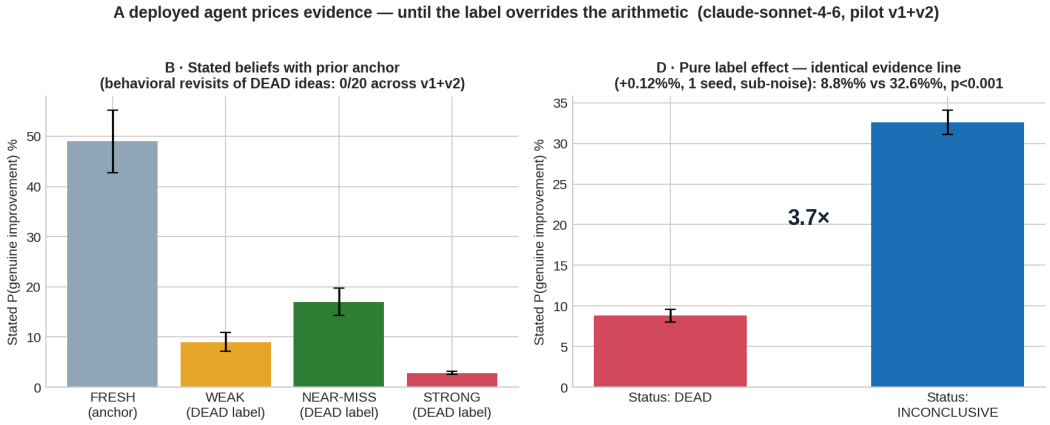


Figure 10: Stated beliefs with a prior anchor (left); the pure label effect on identical evidence (right).

Display name	API model ID	Route	Date
Claude Haiku 4.5	`claude-haiku-4-5`	Anthropic	2026-08-19/20
Claude Sonnet 4.6	`claude-sonnet-4-6`	Anthropic	2026-08-19/20
Claude Opus 4.8	`claude-opus-4-8`	Anthropic	2026-08-19/20
DeepSeek-chat	`deepseek-chat`	DeepSeek	2026-08-20
DeepSeek-R1	`deepseek-reasoner`	DeepSeek	2026-08-20
GPT-4o	`openai/gpt-4o`	OpenRouter	2026-08-20
Gemini-2.5-flash	`google/gemini-2.5-flash`	OpenRouter	2026-08-20
Llama-4-Scout	`meta-llama/llama-4-scout`	OpenRouter	2026-08-20
Mistral-medium-3	`mistralai/mistral-medium-3`	OpenRouter	2026-08-20
Kimi-K2	`moonshotai/kimi-k2`	OpenRouter	2026-08-20
GLM-4.7	`z-ai/glm-4.7`	OpenRouter	2026-08-20
Qwen3-32B	`qwen/qwen3-32b`	OpenRouter	2026-08-20

OpenRouter routes to third-party inference providers, so serving-side configuration for those rows is not under our control; per-call responses are included in the released artifact. Qwen3-32B's belief

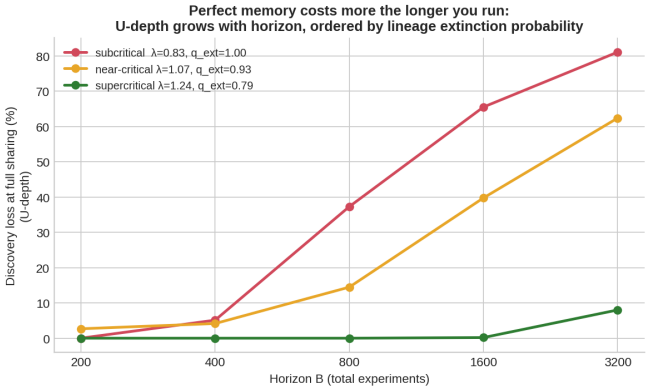


Figure 11: Discovery loss at full sharing grows with horizon, ordered by extinction probability.

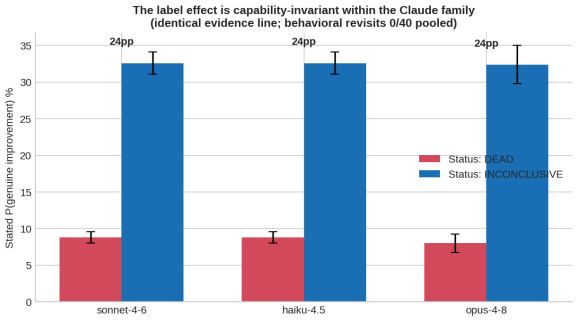


Figure 12: Within the Claude family, the label effect on stated beliefs is undiminished across the tested scale range.

probes returned empty completions under the reasoning cap and are excluded from belief analyses; its choice data are retained. Verbatim prompt templates and the runnable protocol are in the released artifact.

Belief-level taxonomy (descriptive, 5–6 trials per model-condition). Provenance-*unlockable* (Claude; DeepSeek-R1 $1.5\% \rightarrow 50\%$); *hard-locked* despite demonstrated range (GPT-4o $0 \rightarrow 0$, INCONCLUSIVE 31%); *calibrated-but-choice-locked* (GLM-4.7); near-*immune* (Kimi-K2); floor-dominated pessimists whose $\leq 3\%$ anchors leave the provenance test uninformative (Gemini, Mistral). Labels are run-level descriptions, not stable model properties.

Behavioral replication (preregistered). A fresh-run replication (same authors, new base run; see deviation log) ran the F2 and chemistry choice protocols at 12 trials/condition on all nine non-Claude models (PREREG_BX in the artifact). Deviations, disclosed in the shipped log: the extension attached to a fresh base run rather than the original logs (hence reported as an independent replication, never pooled with the original study), and the F3 arm did not execute due to a log-file collision (F3 remains at original n). Per-family disclaimer repair (marked picks, unparseable excluded): R1 $10/12$, Gemini $8/11$, Kimi $4/11$, GLM $3/6$, Llama $2/10$, Mistral $2/12$, DeepSeek-chat $1/12$, GPT-4o $1/12$, Qwen3 $0/9$; plain DEAD pooled $6/91$. Chemistry: $99/99$ under INCONCLUSIVE vs. $18/82$ under DEAD, per-family suppression in $9/9$.

Statistical hierarchy. Calls nest within models and models within families; pooled call-level tests (Fisher, t) treat calls as exchangeable and should be read as descriptive strength. The design-robust headline is the family-level exact sign test ($9/9$ families under DEAD vs. INCONCLUSIVE, two-sided $p = 0.0039$), which is immune to within-family pseudo-replication.

C Proof sketches

Lemma 1 (threshold optimality). Consider the myopic memory decision for a recorded idea with posterior $p = P(\delta \le 0 \mid \text{evidence})$: locking saves an expected retest cost $c > 0$; not locking yields expected value $(1-p)\,V - c$, where $V > 0$ is the expected discounted discovery value of an exonerated idea, treated here as a constant of the environment. Locking is optimal iff $(1-p)V - c \le 0$, i.e. iff $p \ge 1 - c/V \equiv \theta$: a deterministic posterior threshold. Any rule that locks with probability $g(p)$ for non-degenerate g (probability matching, $g(p) = p$) mixes the two actions on the wrong side of the cut and is weakly dominated pointwise, strictly on any posterior mass in $(\theta, 1) \cup (0, \theta)$ where $g \notin \{0, 1\}$. In the full sequential problem the value V of an exonerated idea acquires horizon and lineage terms (children of exonerated ideas); since $\theta = 1 - c/V$ is increasing in V, raising the continuation value *raises* the locking bar: more future value at stake demands more certainty before an irreversible lock. Horizon, however, also inflates the opportunity cost c of spending a slot on a retest (the foregone fresh test has descendants too), so the net movement of θ is model-dependent. We do *not* claim the sequential rule remains a threshold in p alone: if descendant value depends on the magnitude of δ, two records with equal p but different posterior shapes above zero can carry different continuation values, so p need not be a sufficient statistic. The threshold rule we evaluate empirically is the one-step Bayes rule; its end-to-end advantage (Sec. 8) is measured, not derived from a sequential optimality claim. □

Prop. 1. Condition on the binary event $F = \{y \le 0\}$ (possibly repeated). $P(F \mid \delta) = \Phi(-\delta/\sigma)$ is strictly decreasing in δ, so the likelihood of any conjunction of such events has monotone (decreasing) likelihood ratio in δ; hence the posterior given the events is first-order stochastically dominated by the prior. Confirmation probability $q(\delta)$ and confirmed value δ^+ are non-decreasing in δ, and children do not exist in the exogenous-supply model, so the per-step value of revisiting a binary-marked idea is dominated by a fresh draw; a stepwise coupling then preserves dominance of cumulative discovery. □

Remark (why the binary qualifier is necessary). The dominance fails for measurement-valued records: with $\mu_0 = -1$, $\tau = \sigma = 1$, a fresh draw has $P(\delta > 0) = \Phi(-1) \approx 15.9\%$, while after observing $y = -0.01$ (a failure) the posterior is $\delta \mid y \sim \mathcal{N}(-0.505,\, 0.5)$ and $P(\delta > 0 \mid y) \approx 23.8\% > 15.9\%$: a near-null failure is favorable evidence relative to a pessimistic prior. This is precisely why evidence-graded records retain value in exchangeable settings, and why banks that store only DEAD destroy it.

Prop. 2(a). $(\delta, \hat{\delta})$ is bivariate normal with means (μ_0, μ_0), variances $(\tau^2, \tau^2 + \sigma^2)$, covariance τ^2. Standardize and express $P(\delta > 0, \hat{\delta} \le 0)$ as an orthant probability with correlation $-\rho$; divide by $P(\hat{\delta} \le 0)$. Limits follow from $\rho \to 1$ and $\rho \to 0$. (b) is the same computation on $(\delta, X, X+Y_1+Y_2)$ with $\mathrm{corr}(X, S) = 1/\sqrt{3}$. (c) is immediate from the Gaussian c.d.f. □

Thm. 1. *Standing assumptions.* (A1) The type space is a compact interval $[-M, M]$; all statements concern this truncated process, and all reported numerics use truncations whose boundaries lie well outside the simulated quality mass. For $b < 1$ the child-mean map $\delta \mapsto \mu_c + b(\delta - a)$ is a contraction, so quality mass is naturally confined; for $b \ge 1$ the offspring mean can drift outward and we make no claim about the untruncated process. (A2) The mean kernel $Cq(\delta)k(\delta' \mid \delta)$ is jointly continuous and strictly positive on $[-M, M]^2$, and offspring counts are bounded by C. (A3) b, τ_c, σ are finite and positive. Under (A1)–(A2) the mean operator is compact and strongly positive on $C([-M, M])$, so the Krein–Rutman theorem gives a simple leading eigenvalue $\lambda > 0$ with strictly positive eigenfunction h; the spectral radius exists and equals λ. The process is a general-type Galton–Watson process: an individual of type δ produces, with probability $q(\delta)$, exactly C offspring i.i.d. from $k(\cdot|\delta)$, else none; assumptions (A1)–(A2) supply the Krein–Rutman eigenpair (λ, h) as stated. (i) For $\lambda < 1$. Write $Z_n(f) = \sum_i f(\delta_{n,i})$ for the generation-n population functional. By the eigenvalue property, $\mathbb{E}[Z_{n+1}(h) \mid \mathcal{F}_n] = Z_n(Ah) = \lambda Z_n(h)$, so $\mathbb{E}[Z_n(h)] = \lambda^n Z_0(h)$. Since h is continuous and strictly positive on the compact $[-M, M]$, $h_{\min} := \min h > 0$, hence the population size satisfies $|Z_n| \le Z_n(h)/h_{\min}$ and $\mathbb{E}|Z_n| \le \lambda^n Z_0(h)/h_{\min}$. Markov's inequality gives $P(|Z_n| \ge 1) \le \lambda^n Z_0(h)/h_{\min} \to 0$; because $\{|Z_n| \ge 1\}$ is decreasing in n (extinction is absorbing), $P(\text{survival}) = \lim_n P(|Z_n| \ge 1) = 0$: extinction is almost sure. For total progeny, $\mathbb{E}\sum_{n\ge 0}|Z_n| \le \frac{Z_0(h)}{h_{\min}} \sum_{n\ge 0} \lambda^n = \frac{Z_0(h)}{h_{\min}(1-\lambda)} < \infty$, a bound independent of the run budget B; confirmations under full deference are dominated by total progeny, giving the uniform-in-B bound in the statement. (ii) For $\lambda > 1$. The offspring generating functional is

$F[s](\delta) = 1 - q(\delta) + q(\delta)\big(\int s\, dk_\delta\big)^C$ for measurable $s : [-M, M] \to [0, 1]$; the type-δ extinction probability vector $q_{\text{ext}}(\cdot)$ satisfies $q_{\text{ext}} = F[q_{\text{ext}}]$ by first-generation decomposition, and is the *minimal* solution of this fixed-point equation in $[0, 1]^{[-M,M]}$: iterating $s_{k+1} = F[s_k]$ from $s_0 \equiv 0$ gives a pointwise-increasing sequence (monotonicity of F) whose limit solves the equation and lower-bounds any other solution by induction. Strict positivity of the kernel (A2) makes the process irreducible in the multi-type sense, and $\lambda > 1$ then implies $q_{\text{ext}}(\delta) < 1$ for every δ, with survival probability $1 - q_{\text{ext}}(\delta) > 0$ bounded away from zero on compacts; these are the standard general-type statements (Harris 1963, Ch. III; Athreya–Ney 1972, Ch. V), whose proofs carry over verbatim under (A1)–(A2) since the mean operator is compact with simple leading eigenvalue. *Status of the scalar reduction.* The scalar q_{ext} quoted in the main text solves the single-type equation $x = 1 - \bar{q} + \bar{q}\, x^C$ with $\bar{q} = \int q(\delta)\, \pi_h(d\delta)$, where π_h is the normalized left-eigenmeasure pairing of the Krein–Rutman eigenpair: it collapses type structure to an h-weighted average confirmation probability. It is a *numerical organizer*, not a theorem: we prove no error bound between it and the type-dependent minimal solution, and use it only to order the simulated horizon curves, which it does at every horizon tested (Fig. 11). All claims of the theorem proper concern the truncated multi-type process above. □

D Additional analyses (camera-ready)

This appendix answers questions raised in review; every number below is recomputed from the released artifact (`sysloop/truth_sensitivity.py`, `e6/`).

Finite-seed truth labels. Each confirmed config in the end-to-end loop is truth-tested with 6 fresh seeds. Instead of counting a discovery when the estimated effect is positive, we require the improvement to clear a one-sided confidence criterion that propagates both the truth runs' standard error and the baseline's (12 runs). Discoveries per loop (8 replicates; graded minus binary with Welch 95% CI; exact one-sided rank tests as in the main text):

criterion	none	binary	parole	bandit	graded	graded−binary
point estimate (main text)	46.5	46.6	45.8	50.0	60.5	13.9 [9.7, 18.1]
one-sided 80%	44.4	44.6	43.5	46.8	58.1	13.5 [8.0, 19.0]
one-sided 90%	43.5	44.0	42.8	45.8	57.2	13.2 [7.6, 18.9]
one-sided 95%	41.4	42.2	40.8	44.2	56.1	13.9 [7.9, 19.8]

All conditions lose a few marginal discoveries under stricter criteria; the relative margin stays +30–33%, and graded beats none, binary, and parole at $p \le 1.5 \times 10^{-3}$ under every criterion.

What $P(\textbf{dead})$ is computed from. The graded rule sees only the loop's own history: each record's pooled runs (first test plus any verification runs) are compared with the loop's 12-run baseline mean and standard deviation under a flat prior, $P(\text{dead}) = \Phi(z)$ with $z = (\bar{y} - \hat{\mu}_{\text{base}})/(\hat{\sigma}/\sqrt{n})$, and a config is skipped iff $P(\text{dead}) > 0.8$. No effect prior, landscape statistic, or retrospective quantity enters. The threshold is not tuned: a preregistered sweep over $\theta^* \in \{0.6, 0.7, 0.9, 0.95\}$ gives 63.6/63.4/61.6/62.1 discoveries, all above binary at $p \le 5 \times 10^{-4}$, with the recorded θ^*=0.8 (60.5) within noise of all four. We have not computed a sequential optimum; the sweep shows the one-step rule's advantage is not a knife-edge.

The low-noise cell. $\sigma/\tau = 0.24$, budget 80, four memory rules on the same cached noise realizations, 4 paired replicates: binary 18.0, parole 17.5, threshold 16.0, none 14.2. It was run before the noisy cell, at a location the simulated phase diagram places on the monotone side of the boundary, but it was not preregistered and four replicates do not support a precise difference; we report its sign only.

Second proposer family in the enforced gated cell. Preregistered (PREREG_E6), GPT-4o proposer, same landscape and harness, 12 replicates. Soft lock: none 3.7 ± 2.2, binary 9.1 ± 2.3, graded 10.1 ± 2.9. Enforced: binary 13.4 ± 4.6, graded 12.2 ± 1.7; the preregistered direction (graded > binary) is not supported ($p = 0.96$). The mechanism matches DeepSeek: strict rescues under enforced binary 0, 54 wrongfully locked good ideas, 418 blocked retries; graded rescues 24 through OPEN records. Without a bank GPT-4o re-proposes so heavily (12–25 distinct configs per 150-step loop) that the lock's efficiency gain outweighs its fertility loss at this budget.

Beyond Gaussian effects. The replays use each loop's empirical effect distribution, not the Gaussian plug-in, which is why the Gaussian prior underpredicts there. The small-noise behaviour of the first-test rate does not depend on the Gaussian form: for any bounded effect density f continuous at 0 with $P(\delta < 0) > 0$, $W_A \sim \sigma f(0^+)/\big(\sqrt{2\pi}\, P(\delta < 0)\big)$ as $\sigma \to 0$ (substitute $\delta = \sigma u$ in $\int_0^\infty \Phi(-\delta/\sigma) f(\delta)\, d\delta$ and apply dominated convergence; $\int_0^\infty \Phi(-u)\, du = 1/\sqrt{2\pi}$). The density at the decision boundary, not the effect variance, sets the rate—the regularity behind FCNet's low aggregate rate despite 44% near-marginal convictions.

Positive-side exposure. We measure that false confirmations draw descendant tests whose confirm rate falls ($75\% \to 59\%$), but not whether a pipeline converts that into retiring the record within a finite window w. The end-to-end loop never retires a confirmed record, so its result does not depend on positive-side correction; the inverted U in simulation does (Fig. 6B).